# The closing corridor: postponement against the effective ceiling of the fertility lifecourse

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**Abstract** Whether a cohort can reach replacement fertility depends on when childbearing starts and when it must end. Ages at first birth have risen five to six years since the 1970s; this paper is about the other boundary. Using the updated 45-country HFD/HFC panel (1850–2025), we define the effective ceiling of the fertility lifecourse as the highest age with ASFR ≥ 1/200 and show it is behavioural, not biological: 47–48 in nineteenth-century populations, pulled down to 41 by the 1980s as parity control truncated high-parity childbearing, and recovered only to about 43 since — one year per fifteen calendar years. The fertility mass above age 40 (0.02–0.07 children) cannot arithmetically substitute for earlier childbearing. Consequently replacement must now be achieved within a corridor roughly thirteen years wide, and on composite lattice plots the replacement contour visibly terminates at the ceiling — distinguishable from right-censoring — in 43 of 45 countries' recent cohorts. The corridor framing contributes discipline to the debate on accelerating fertility decline: any candidate explanation must match the same age–cohort–timing fingerprint, and postponement converts to quantum loss even under unchanged intentions.

**ℹ Note**

**Draft status.** Early working version posted for open comment. Reproducible from https://github.com/JonMinton/Comparative_Fertility; analytic choices (thresholds, definitions) were fixed in the repository's workplan before this analysis ran on the updated data, with prototypes on the pre-update data disclosed there.

## Introduction

Two quantities jointly determine whether a cohort can reach replacement fertility: when childbearing starts, and when it must end. The first has moved enormously — mean ages at first birth have risen five to six years since the 1970s across the low-fertility world [1], [2]. This paper is about the second: the *effective ceiling* of the fertility lifecourse, its history, and its consequence — a corridor between a rising floor and a nearly fixed ceiling within which replacement must now be achieved.

We make three claims. First, the ceiling is behavioural, not biological: in the nineteenth-century populations of our panel it sat at ages 47–48, was pulled down to 41 by the 1980s as parity control truncated high-parity childbearing, and has since crept back only to about 43. Second, the fertility mass above age 40 is small enough (0.02–0.07 children) that no plausible late-fertility recuperation can substitute for earlier childbearing [2]. Third, the interaction of these facts with cumulative cohort fertility is directly visible on composite lattice plots [3]: the replacement contour *goes vertical* where it meets the ceiling, and the distinction between contours that hit the wall and contours that are merely right-censored is itself informative.

## Data and methods

Data: the updated 45-country HFD/HFC panel described in the companion descriptive finding (combination rules of S. Pattaro, L. Vanderbloemen, and J. Minton [3] applied to the 2026 releases [4], [5]; 1850–2025).

Definitions, fixed ex ante in the public workplan:

- **Effective ceiling** of population $c$ in year $t$: the highest age at which ASFR $\geq$ 1/200 (primary threshold), with 1/1000 as sensitivity.
- **Late-fertility mass**: cumulative ASFR at ages 40+ (and 35+) in a given country-year.
- **Reachability**: for each cohort observed from age 16 or earlier, the age at which CPCFR first reaches 2.05, classified as *crossed*; *never* (observed beyond age 44 without crossing); or *censored*.

## Results

### The ceiling is behavioural and nearly fixed

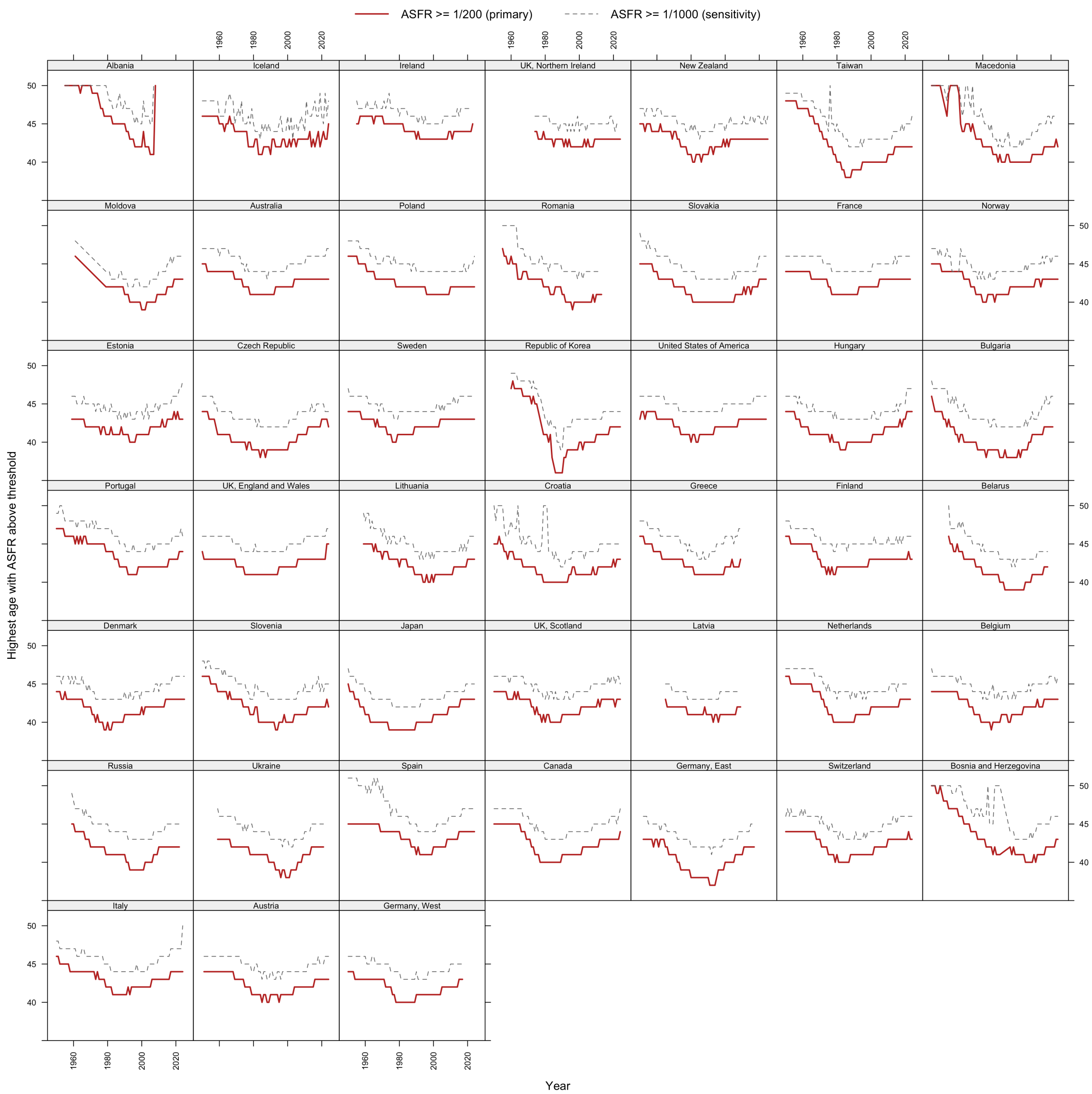


Figure 1: Effective ceiling trajectories by country, 1950–2025: highest age with ASFR at or above 1/200 (solid) and 1/1000 (dashed).

Pooled decade medians of the primary ceiling: 45 (1950s), 44 (1960s), 42 (1970s), 41 (1980s and 1990s), 42 (2000s), 42 (2010s), 43 (2020s). The recovery since the 1990s trough amounts to roughly one year of age per fifteen calendar years (Figure 1).

## The mass behind the ceiling is small

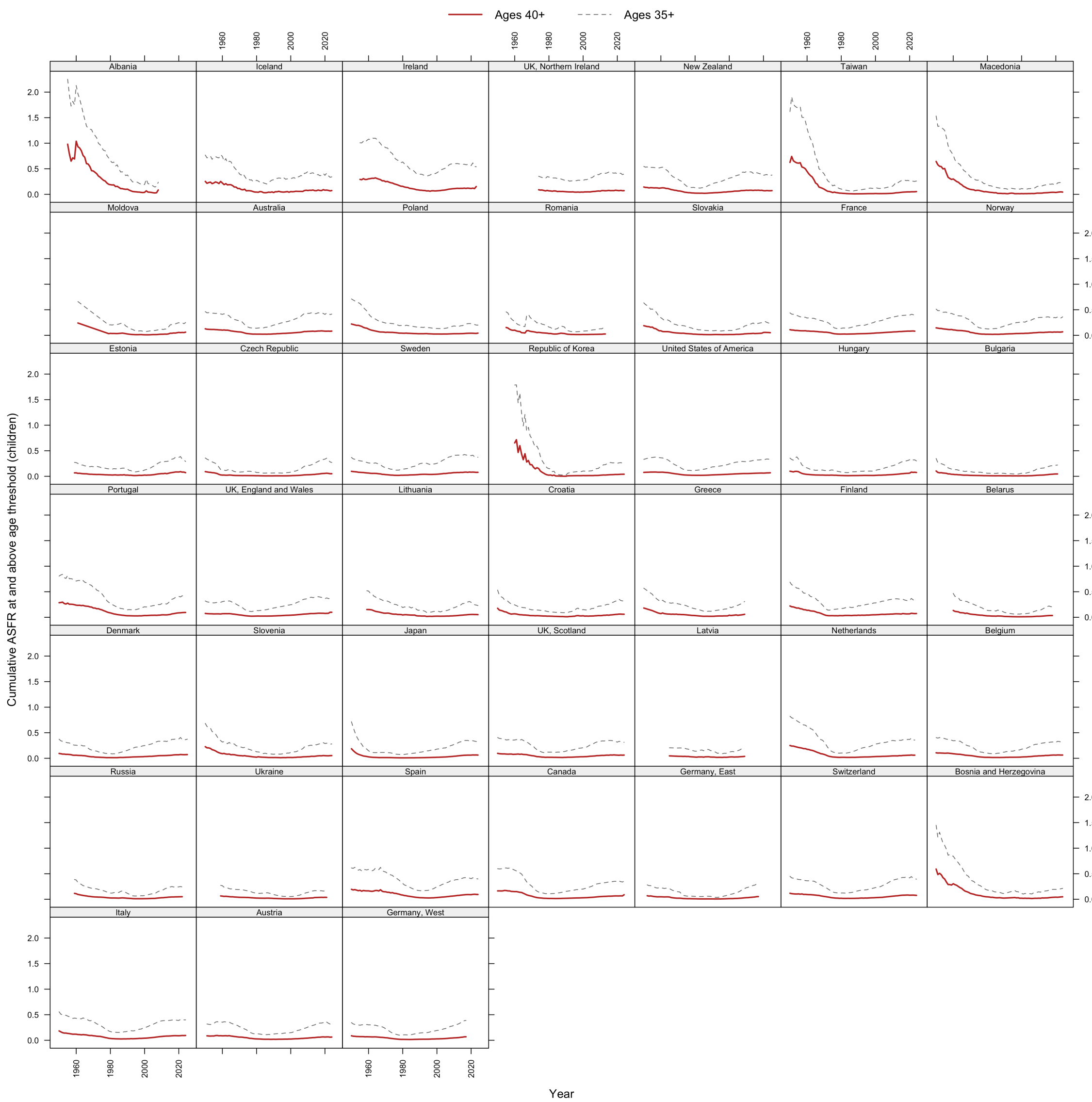


Figure 2: Cumulative ASFR at ages 40+ (solid) and 35+ (dashed), by country, 1950–2025.

Median cumulative fertility above age 40 fell from 0.117 (1950s) to 0.023 (1980s) and has recovered to 0.065 (2020s) (Figure 2) — at its twenty-first-century maximum, about 3 percent of a replacement target. The 40+ segment cannot arithmetically rescue replacement for cohorts entering their forties below roughly 1.99.

## Reachability: where the replacement contour meets the wall

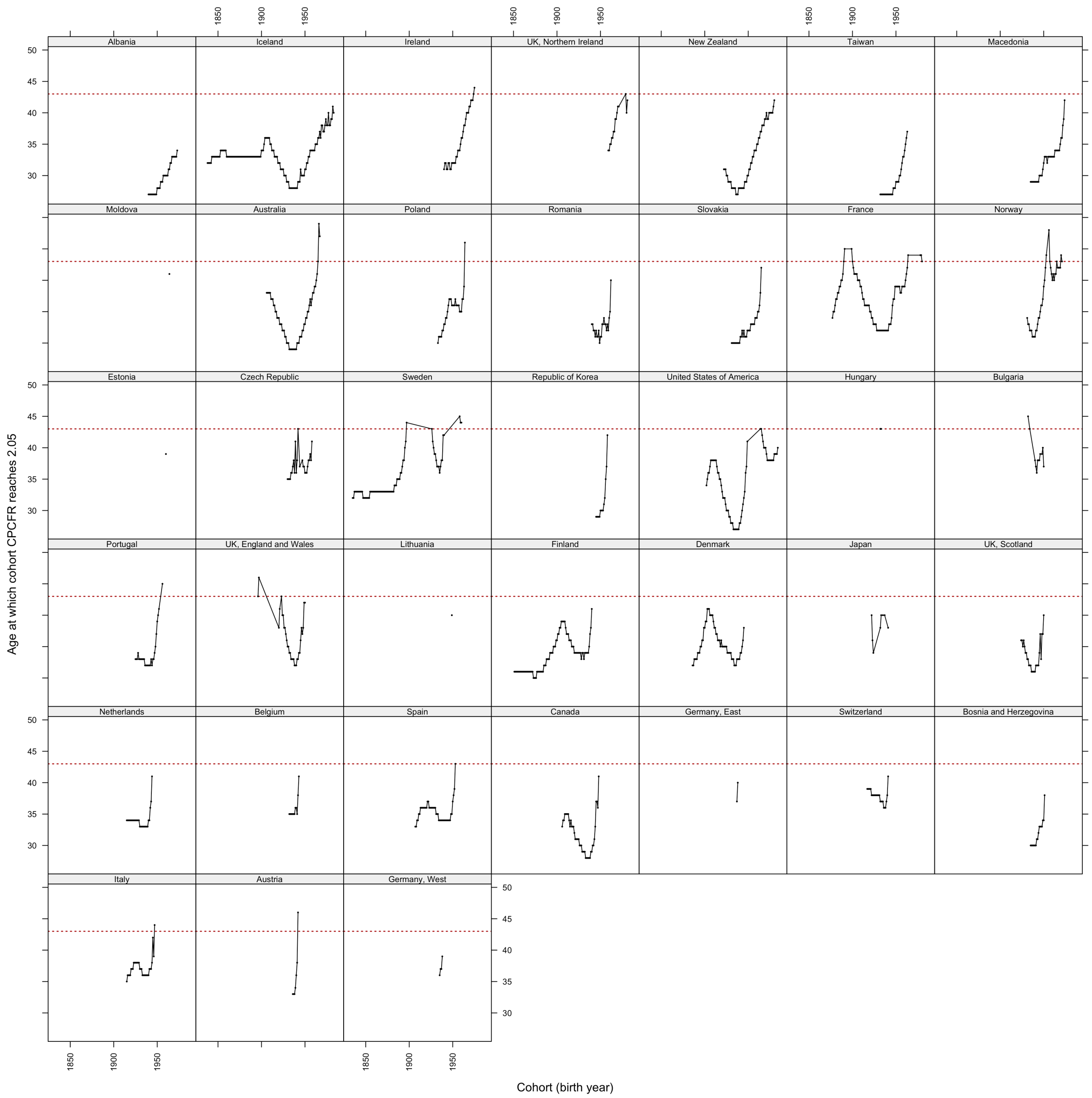


Figure 3: Age at which each cohort's CPCFR reaches 2.05, by country; the dotted reference line marks age 43.

Forty-three of 45 countries have post-war cohorts observed beyond age 44 that never reached 2.05; only nine have any cohort born after 1970 that crossed. Crossing ages drift upward toward the ceiling before crossings cease (Figure 3).

## The wall on the surface

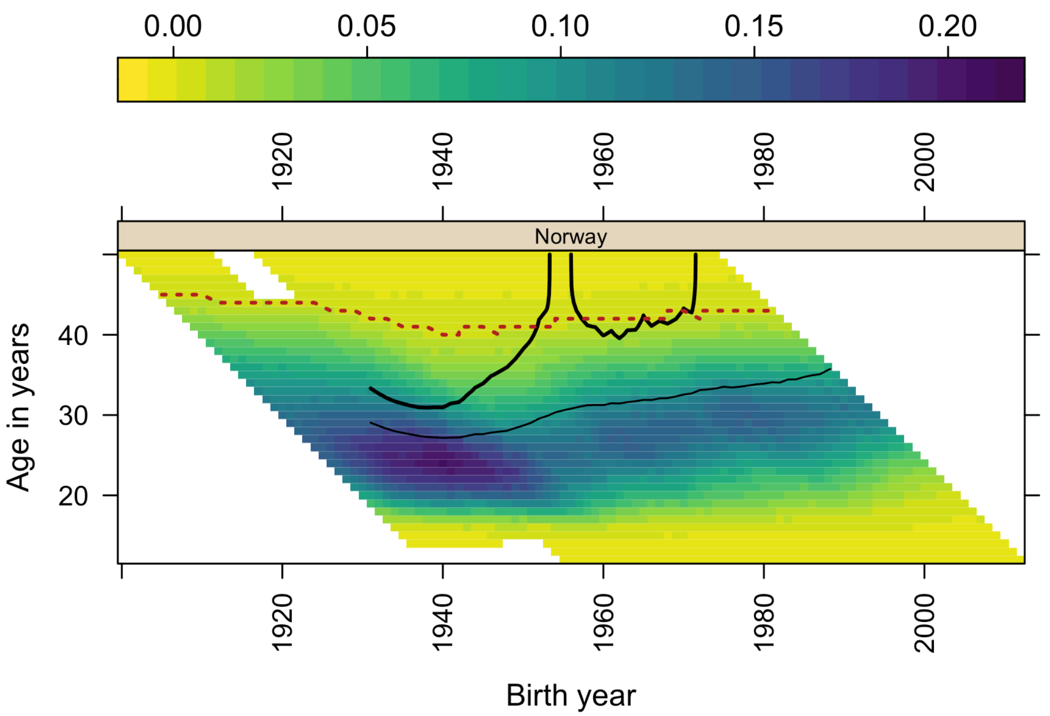


Figure 4: Norway: composite surface with the effective ceiling overdrawn (dotted). Both vertical escapes of the 2.05 contour terminate at the ceiling.

Overlaying the ceiling on the composite surface (Figure 4; US and South Korean panels in the repository) makes the two contour-termination modes visually distinct: contours that climb into the ceiling and end there (wall-terminated) versus contours cut by the right edge of the data (censored). For Norway, both escapes of the replacement contour are wall-terminations, not censoring artifacts.

A note on visual complexity: in grammar-of-graphics terms the ceiling adds a fifth encoding to an already four-dimensional composite (x, y, shading, contour weight) — though it is better understood as a third *reading* of the single underlying ASFR field: its level (shading), its accumulation (contours), and its effective boundary (the ceiling), each a computation the eye cannot perform unaided. Because four encodings already approach the limit of comfortable readability for many audiences, we confine this densest form to the present paper; the companion descriptive finding retains the established four-encoding composite throughout.

## Discussion

The corridor framing contributes discipline, not adjudication, to the current debate on accelerating fertility decline. Any candidate explanation — economic precarity [6], [7], housing, partnering, smartphone-era social change — must match the same fingerprint: *which ages* lost fertility, *which cohorts* carry the loss, whether milestones are delayed (tempo) or abandoned (quantum), and why near-identical exposures produce heterogeneous national responses [8], [9]. The composite surfaces render that fingerprint directly; they cannot, and we do not, pick among co-timed period candidates. What the ceiling adds is arithmetic: postponement is not neutral even under unchanged intentions, because trajectories that start later run into an upper boundary that has moved two years while starting ages moved six. Replacement has become a corridor roughly thirteen years wide.

Limitations: assisted reproduction is shifting the ceiling slowly and unevenly; ceiling estimates at thresholds are sensitive to small-population noise (hence the per-country trajectories as primary output); the cohorts most relevant to the current debate remain right-censored; and single-year ASFR data cannot separate parity-specific dynamics — the natural extension, using the by-birth-order collections already assembled in the repository [10].

## References

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